



Review

Left atrial appendage closure for stroke prevention in nonvalvular atrial fibrillation: A current overview

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ABSTRACT

Atrial fibrillation (AF) is the most common arrhythmia encountered in clinical practice and its prevalence increases with age. AF causes palpitations, heart failure, and cardiogenic embolism. Of them, the most critical and fatal complication is a cardio-embolic event. Oral anticoagulation plays a central role in reducing stroke risk in AF patients. Recently, when oral anticoagulation is considered in patients with non-valvular AF who are eligible for direct oral coagulants, they are preferred to vitamin K antagonist based on accumulating evidence. Although many patients can tolerate oral anticoagulation, there is a subset of patients who cannot tolerate long-term oral anticoagulation. Such a subset has a higher bleeding risk as indicated by the HAS-BLED score under oral anticoagulation. This subset of patients requires effective and safe non-pharmacological alternative therapies for stroke prevention. One of the promising non-pharmacological therapies is left atrial appendage closure. Three randomized controlled trials demonstrated non-inferiority of percutaneous left atrial appendage closure using WATCHMAN family to oral anticoagulation (Boston Scientific, Marlborough, MA, USA). WATCHMAN FLX, which was innovated following WATCHMAN 2.5, was associated with fewer safety events and a higher success rate of effective appendage closure. Nevertheless, several unsolved issues remain, including device-related thrombosis, post-treatment antithrombotic therapy, and peri-device leakage. Left atrial appendage closure for patients with non-valvular AF may be an alternative therapy to avoid cardiac embolism for high bleeding risk patients with contraindications to long-term oral anticoagulation therapy.

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Introduction

Atrial fibrillation (AF) is the most common arrhythmia encountered in clinical practice and its prevalence increases with age. In Japan, an epidemiological study was conducted by the Japanese Circulation Society [1]. The percentage of AF increased up to 3.44 % for men and 1.12 % for women in their 70s; 4.43 % for men and 2.19 % for women over 80 years old. When calculated using future population projections, it is predicted that by 2050 there will be approximately 1.03 million AF patients, accounting for approximately 1.1 % of the total population.

One of the most critical and fatal complications of AF is a cardioembolic event. AF is associated with a 4- to 5-fold incremental risk of ischemic stroke [2].

Oral anticoagulation (OAC) plays a central role in reducing stroke risk in AF patients. Randomized controlled clinical trials of vitamin K antagonists (VKA) and direct oral anticoagulants (DOACs) have established OACs as a standard of care for stroke prevention in the AF cohorts [3].

Recently, when OACs are considered in patients with non-valvular AF (NVA) who are eligible for DOACs, DOACs are preferred to VKA based on accumulating evidence (level of evidence IA) [4].

Although many patients can tolerate OACs, there is a subset of patients who cannot tolerate long-term OACs even with DOACs. Such a subset has a higher bleeding risk as indicated by the HAS-BLED score under OACs. HAS-BLED score is calculated from (1) uncontrolled hypertension, (2) abnormal renal/liver function, (3) stroke history, (4) bleeding history, (5) labile INR, (6) elderly, and (7) drug/alcohol [5].

This subset of patients requires effective and safe nonpharmacological therapies for stroke prevention. One of the promising nonpharmacological therapies is left atrial appendage closure (LAAC). In this review, we

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describe a summary of LAAC which is performed to prevent stroke in patients with NVAf and high bleeding risks.

LAA anatomy and function

LAA anatomy

Wang and colleagues [6] presented the LAA has a wide variability in its size and shape. The LAA ostium is oval-shaped in most cases (69 %), otherwise round, water drop, or foot-like. The LAA size is the following measurements: mean LAA ostium dimensions of 25.4 ± 5.5 mm in maximum and 16.8 ± 4.5 mm in minimum in Westerners. Japanese have larger LAA ostium than non-Asian populations [7].

The body of LAA is trabeculated with pectinate muscles. LAA morphology is classified into four types, which were assessed using multi-detector computed tomography and cardiac magnetic resonance. Multi-detector computed tomography is superior in evaluating LAA morphology to trans-esophageal echocardiography [8] (Fig. 1). The predominant shape is “chicken wing” (48 %), followed by “cactus” (30 %), “windsock” (19 %), and “cauliflower” (3 %) [9].

A basic comparison of anatomic reports shows that the most common LAA shape differs by geography. For example, cactus predominates in the Japanese population [10]. Of note, the “cauliflower” morphology, which has a variable number of lobes, is most often associated with embolic events. There is a strong relationship between the number of lobes and the incidence of LAA thrombus. For example, thrombus was observed only in 0.7 % in patients with AF having 1 or 2 lobes [11].

LAA function

LAA has a hemodynamic function as a reservoir. LAA has another important role in the secretion of neuro-hormonal factors, including natriuretic peptides: atrial natriuretic peptide (ANP) and B-type natriuretic peptide (BNP) [12].

ANP has a key role in the maintenance of systemic fluid balance. Surgical resection of LAA might decrease the level of ANP secretion and facilitate volume overload [13]. Majunke and colleagues evaluated the trend in plasma ANP levels following percutaneous LAAC [14]. The plasma ANP levels increased immediately following LAAC procedures, accompanied by a decline in plasma ANP levels several days later. The trend in plasma ANP levels varies in each report, but such a trend might not affect patients' hemodynamics [15]. Neuro-hormonal impact of LAAC requires further studies.

How to close LAA, surgical or percutaneous?

Surgical LAAC

LAA thrombus is identified as the source of AF-related strokes in >90 % of cases, whereas thrombus is observed in the LAA in 27 % of all comers with untreated AF [16,17].

Initial attempts to surgically exclude the LAA were via a direct surgical approach usually concurrent to other cardiac surgery for patients with a high risk of stroke. Different techniques of LAA ligation or excision have been proposed. Current surgical techniques to exclude the

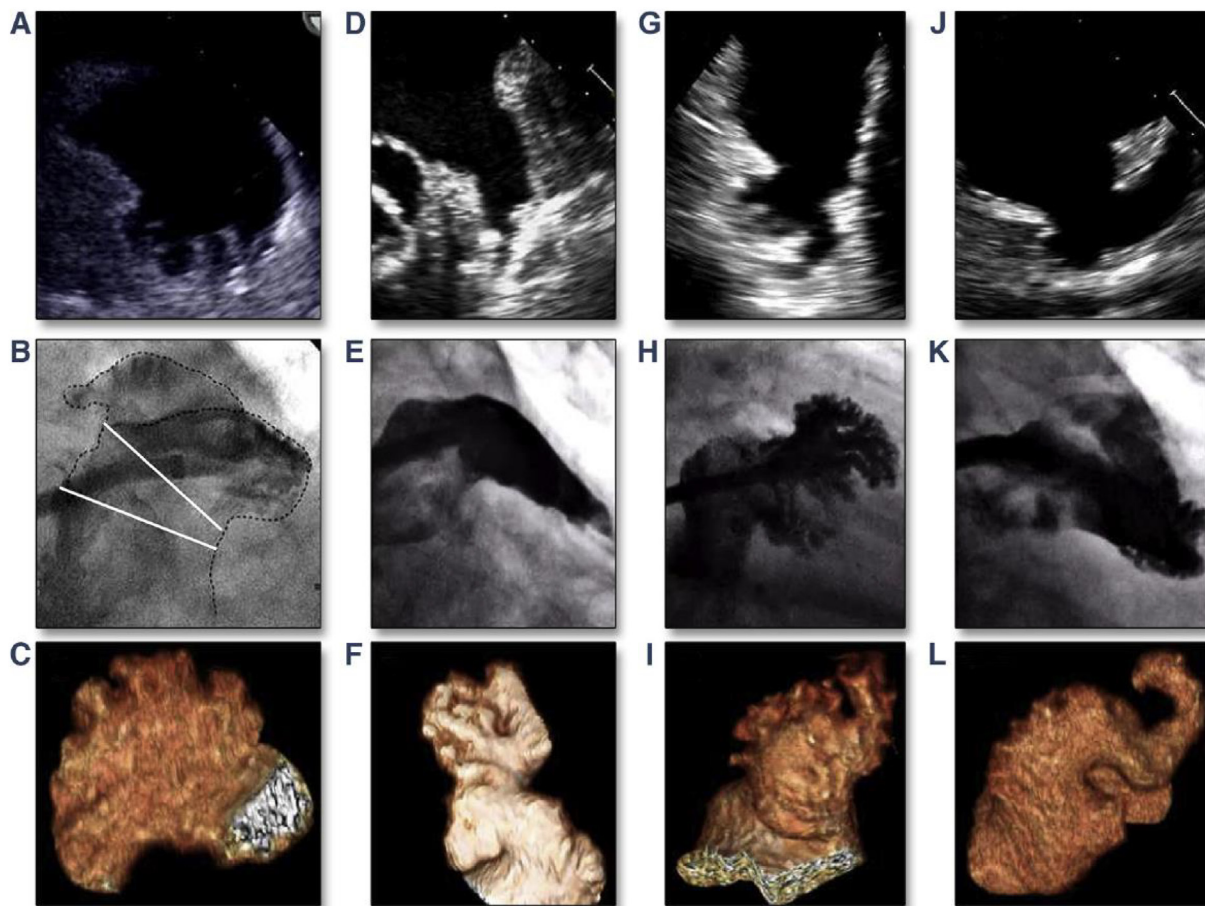


Fig. 1. LAA morphologies and assessment modalities. The four different LAA morphologies as shown by TEE (top), cine angiography (middle), and multi-detector computed tomography (bottom): Cauliflower (A to C), windsock (D to F), cactus (G to I), and chicken wing (J to L). LAA, left atrial appendage; TEE, trans-esophageal echocardiography.

LAA consist of resection, epicardial stapling, clip application, or endoatrial suture closure [18]. The risk-benefit assessment of surgical LAAC is clouded by potential increases in postoperative complications, incomplete appendage closure (19 %–59 %), thrombus formation in the remaining stump (25 %), and subsequent embolic events (2.6 %–8 %) [19–21].

Because of these limited data on surgical LAAC, the 2017 Society of Thoracic Surgeons indicates a Class IIa recommendation for the excision or exclusion of the LAA in undergoing cardiac surgery for another reason in patients with AF [18]. The LAAOS III trial [22] may change the recommendation for surgical LAAC in the future. A total 4770 patients with AF and CHA₂DS₂-VASc₂ ≥ 2 undergoing cardiac surgery for another reason were randomized to receive concomitant surgical LAAC or not. All patients received OACs for the usual postoperative care. At 3.8-years of follow-up, stroke or systemic embolism occurred in 4.8 % in the LAAC group versus 7.0 % in the standard care group. The benefit of surgical LAAC on stroke prevention appeared to be an adjunct to OACs. However, this study does not support surgical LAAC as a replacement for OACs. No study compared surgical LAAC and OACs.

Percutaneous LAAC

In most NVAF patients who do not indicate another surgical indication, the percutaneous approach should have an advantage over the surgical one. OACs have a disadvantage in incremental risk of bleeding. Several percutaneous LAAC devices have been proposed. The WATCHMAN 2.5 and WATCHMAN FLX (Boston Scientific, Marlborough, MA, USA) (Fig. 2A, B) are now available in Japan. Clinical trials using the AMPLATZER AMULET LAA Occluder (Abbott, Santa Clara, CA, USA) (Fig. 2C) are ongoing in Japan.

In contrast to surgical LAAC, three randomized controlled trials are using the WATCHMAN family that compare percutaneous LAAC to warfarin (PROTECT AF and PREVAIL) [23,24] or DOACs (PRAGUE-17) [25].

From 2005 to 2008, PROTECT AF trial was conducted to clarify the safety and efficacy of LAAC using the WATCHMAN. This trial randomized 707 patients with CHADS₂-score [26] ≥ 1 to LAAC using the WATCHMAN2.5 or warfarin (2:1, respectively). For the primary composite endpoint of stroke and systemic embolization and cardiovascular death, LAAC demonstrated non-inferiority (probability >0.999) and superiority (probability >0.960) compared to the warfarin arm. This result was driven by a significant reduction in hemorrhagic stroke (WATCHMAN: 0.6 % per 100 patient-year vs warfarin: 4 % per 100 patient-year) and cardiovascular death (WATCHMAN: 1 % per 100 patient-year vs 2.4 % per 100 patient-year). Although ischemic stroke did not reach non-inferiority, LAAC significantly reduced fatal or disabling strokes (WATCHMAN: 0.5 % per patient-year vs 1.2 % per patient-year).

Following the PROTECT AF trial, the US Food and Drug Administration stated concern that there was a high early safety event rate

including cardiac tamponade, pericardial effusion, and procedure-related strokes.

The PREVAIL trial [24] was conducted specifically to further evaluate the safety endpoint of the WATCHMAN2.5 from 2010 to 2012. PREVAIL trial randomized 407 patients with CHADS₂ ≥ 2 to WATCHMAN2.5 arm and warfarin arm (2:1, respectively). In contrast to PROTECT AF trial, LAAC did not reach non-inferiority for the composite endpoint of stroke and systemic embolization and cardiovascular death. There was an unusually low ischemic stroke rate in the warfarin arm (0.73 %).

Meta-analysis of patients in PROTECT AF and PREVAIL followed for 5 years demonstrated that the composite endpoint between LAAC and warfarin was similar, whereas LAAC demonstrated 80 % relative risk reduction in hemorrhagic stroke (0.17 % vs 0.87 %, $p = 0.0022$). Of these strokes, 55 % fewer were disabling or fatal. Of note, post-procedure bleeding and even all-cause death favored LAAC [27].

In Japan, the SALUTE study evaluated the safety and efficacy of the LAAC with WATCHMAN2.5 to prevent stroke in patients with NVAF [28]. This trial demonstrated the safety and efficacy of the LAAC with WATCHMAN2.5 for Japanese patients with NVAF, similarly to the findings of PROTECT AF and PREVAIL trials.

Recently, DOACs have become the first-line OAC in real-world practice. Therefore, previous PROTECT AF and PREVAIL data versus warfarin cannot be generalized to contemporary OAC use. PRAGUE-17 trial evaluated the safety and efficacy of LAAC against DOACs, instead of warfarin, from 2015 to 2019 [29]. This study randomized 402 patients with NVAF with a history of cardiac embolism, clinically-relevant bleeding, or satisfying both CHA₂DS₂-VASc ≥ 3 and HAS-BLED ≥ 2. The implanted devices were AMULET, WATCHMAN2.5, or WATCHMAN FLX in 61.3 %, 35.9 %, or 2.8 %. As a post-procedure antithrombotic regimen, most patients received dual anti-platelet therapy (DAPT) with aspirin and clopidogrel for three months, followed by aspirin only. In the DOAC group, the most frequently used anti-coagulant was apixaban. The primary composite endpoint combined both efficacy and safety outcomes: all stroke, systemic embolism, major bleeding, cardiovascular death, and device-related complications.

The primary outcome occurred in 8.6 events per 100 patient-year with LAAC compared with 11.9 events per 100 patient-year with DOACs, achieving non-inferiority. The incidence of nonprocedural clinically relevant bleeding was significantly lower in the LAAC group (3.4 %) than in the DOACs group (5.9 %) with a hazard ratio: 0.55 (95 % confidence interval: 0.31–0.97, $p = 0.039$). Among NVAF patients with a high risk for stroke and bleeding, the non-inferiority of LAAC to DOACs was maintained during long-term follow-up. The rate of non-procedural bleeding events was significantly reduced with LAAC compared with DOAC therapy.

The WATCHMAN FLX was investigated in a prospective non-randomized multicenter trial to clarify its efficacy and safety. PINNACLE FLX study [30] is a single-arm investigational trial that

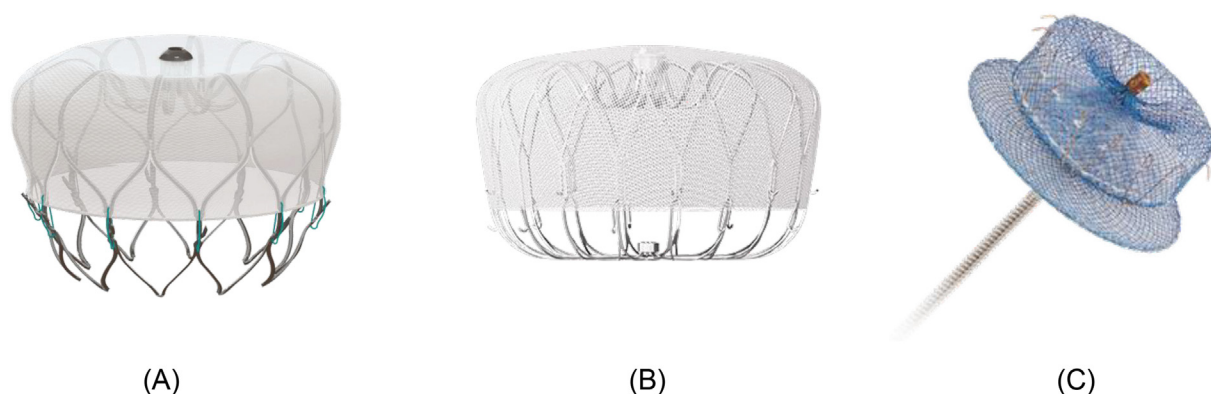


Fig. 2. Percutaneous left atrial appendage closure devices. (A) The WATCHMAN 2.5; (B) the WATCHMAN FLX; (C) the Amplatzer Amlet LAA Occluder.

enrolled 400 patients. The primary efficacy was the incidence of effective LAA closure (peri-device flow ≤ 5 mm), as assessed by the trans-esophageal echocardiography at a 12-month follow-up. The primary safety endpoint was the occurrence of one of the following events within seven days after LAAC: death, ischemic stroke, systemic embolism, and device- or procedure-related events requiring cardiac surgery.

Implant success was 98.9 %. One-year safety events showed 1 % tamponade without requiring surgical repair, 0.5 % procedure-related stroke, with no reported device embolization, and no procedure-related death. This study demonstrated that WATCHMAN FLX is associated with a low incidence of safety events and a high incidence of effective appendage closure.

Percutaneous LAAC with WATCHMAN family

The first-in-human percutaneous LAAC using WATCHMAN2.5 was performed in August 2002, in Europe. The WATCHMAN2.5 received a CE mark in 2005 and Food and Drug Administration approval in 2015. The WATCHMAN FLX, which is a new generation of the WATCHMAN2.5 is approved not only in Europe (March 2019), in the USA (July 2020), but also in Japan (Dec 2020).

The WATCHMAN FLX consists of a self-expanding nitinol 18-struts frame covered with a 160 μ m polyethylene terephthalate fabric mesh cap. This membrane is designed to prevent the embolization of thrombus from the LAA and to facilitate endothelialization. The WATCHMAN FLX is implanted percutaneously to seal the LAA. There are two rows of J-anchors, which fix the WATCHMAN FLX at the ostium of LAA. These anchors are designed so that the WATCHMAN FLX is fully retrievable and can be redeployed without damaging the anchors. There is an atraumatic closure at distal end of the WATCHMAN FLX. This closed structure can be safely placed without damaging the LAA wall. The threaded insert used to connect the delivery system is embedded inside the device to reduce exposed metal to the left atrium. Five sizes are available (20, 24, 27, 31, and 35 mm). The device size should be larger

than the diameter of the LAA ostium to ensure device stability and sufficient compression (10 %–30 %) against the LAA wall (Figs. 3 and 4).

How to implant WATCHMAN?

Planning of LAAC

The use of multimodality imaging is crucial for optimal selection of patient, device, procedure monitoring, and subsequent follow-up. This can be accomplished by trans-esophageal echocardiography or cardiac computed tomography. Trans-esophageal echocardiography is currently considered the gold standard in LAAC pre-procedural assessment and intra-procedural imaging. Recently, computed tomography has been increasingly used to evaluate the morphology and size of LAA pre-procedurally.

Trans-esophageal echocardiography

The presence of LAA thrombus, determination of LAA dimensions and morphology, and feasibility of LAAC should be assessed by the pre-procedural imaging. Trans-esophageal echocardiography can detect thrombus in LAA with high sensitivity (92 %) and specificity (98 %). It has a negative predictive value reaching 100 % and a positive predictive value of 86 % [31].

Detailed assessments of LAA morphology are performed using the 2-dimensional and 3-dimensional trans-esophageal echocardiography. LAA is imaged at 0, 45, 90, and 135° to obtain the LAA measurements and determine the landing zone, maximum diameter (measured from the left circumflex artery to 1–2 cm within the pulmonary vein ridge), and LAA depth (from ostium to the tip of LAA). A multi-planar reconstruction by 3-dimensional trans-esophageal echocardiography allows for precise measurement of the landing zone diameter. The largest LAA landing zone diameter is used for device sizing. The LAA is an even more complex structure, and the true LAA diameter may still be underestimated if measured by trans-esophageal echocardiography alone.

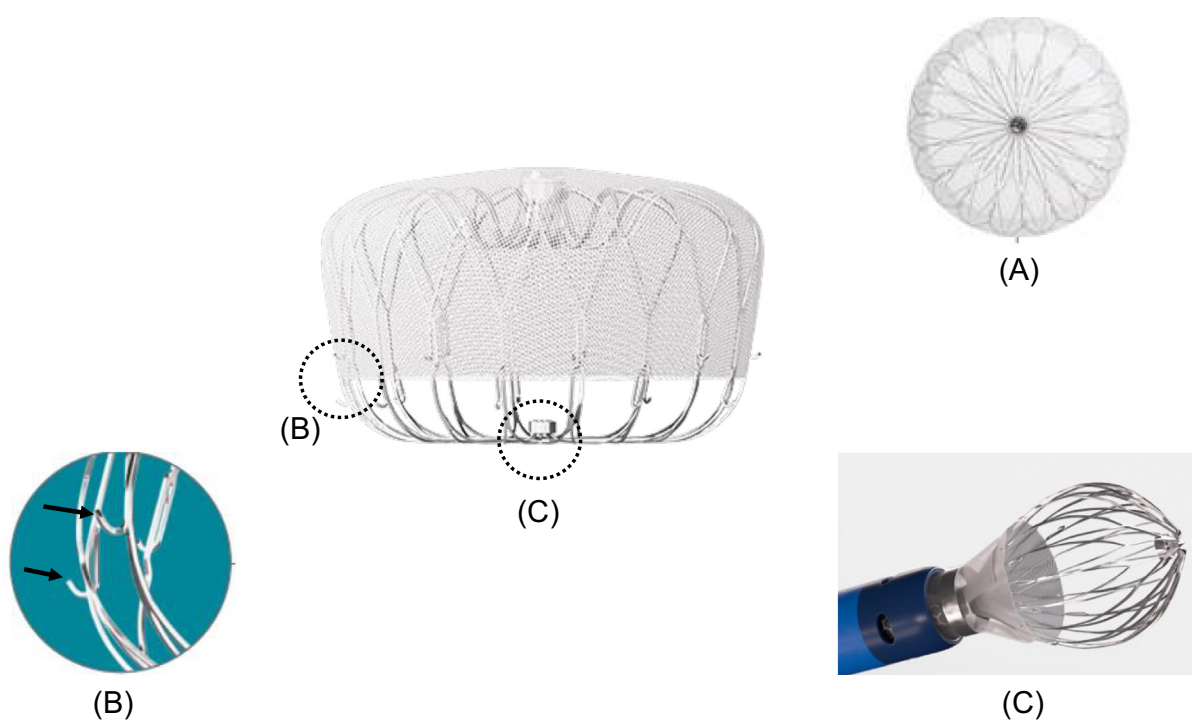


Fig. 3. WATCHMAN FLX. (A) Self-expanding nitinol 18-struts frame covered with a 160 μ m polyethylene terephthalate fabric mesh cap. (B) Two rows of J-shaped anchor. (C) Closet distal end.

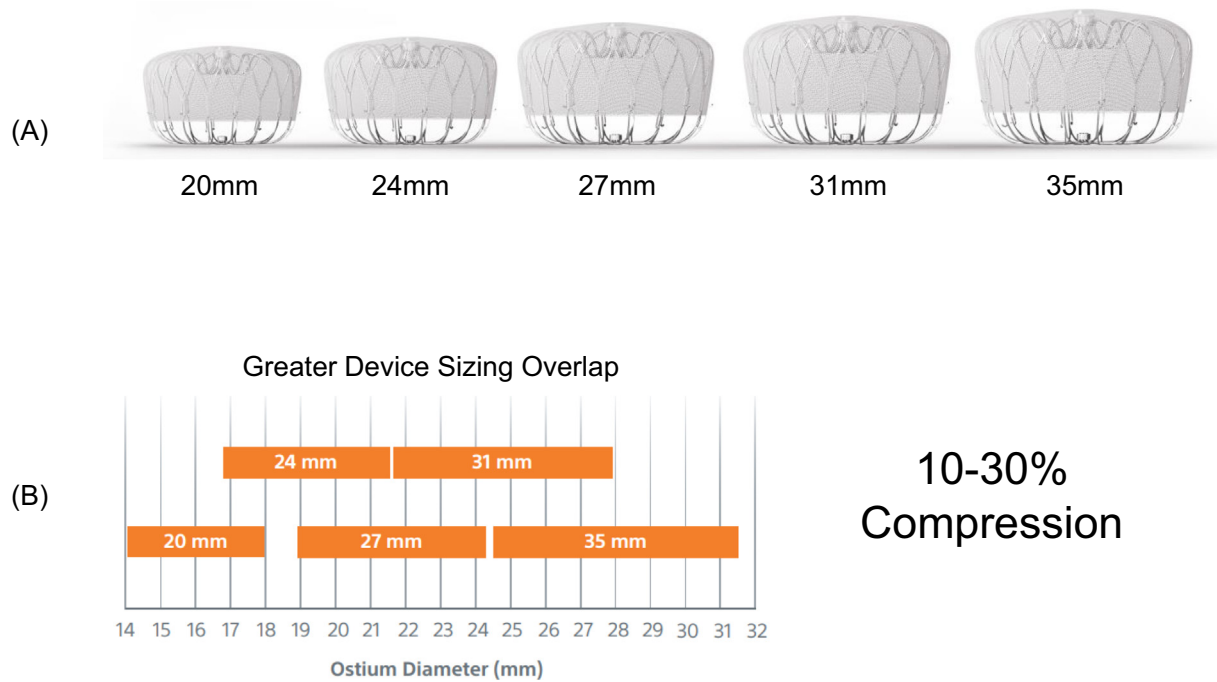


Fig. 4. A variety of sizes of WATCHMAN FLX. (A) Five sizes of the WATCHMAN FLX. (B) Device sizing chart.

Computed tomography

A multi-detector computed tomography generates 3-dimensional volumetric data of the LAA and provides an accurate anatomical assessment. A standard computed tomography analysis includes an assessment of the LAA anatomy in the sagittal, axial, and coronal planes. The perimeter, maximum, and minimum diameters of the LAA landing zone are measured. The existence of LAA thrombus may be evaluated by computed tomography. A computed tomography scan in the prone position is also useful to assess the presence of thrombus within the LAA.

Computed tomography has a high sensitivity (96 %) and a high negative predictive value (100 %) for the evaluation of thrombotic formations [32]. Saw and colleagues compared several modalities including computed tomography, trans-esophageal echocardiography, and fluoroscopy of LAA measurement. They demonstrated that computed tomography was superior to the other modalities in sizing the superior spatial resolution [33].

Procedure techniques

The procedure is performed under general anesthesia and guided by fluoroscopy and 3-dimensional trans-esophageal echocardiography. The device delivery sheath is inserted via a right femoral vein, instead of left femoral vein, because it facilitates trans-septal access. Hemostasis is selected commonly as “figure of eight” technique (Fig. 5) [34] or pre-closure technique using the Perclose ProStyle (suture-mediated closure system, Abbott Vascular, Temecula, CA, USA) (Fig. 6), instead of manual compression. The trans-septal puncture should be at the infero-posterior segment of the fossa ovalis under well-controlled coagulation with activated coagulation time >250–300 s.

Once trans-septal access is established, trans-septal puncture sheath is exchanged to the 14-Fr. device access sheath. The access sheath is available in two types: single or double curve in Japan. The double curve sheath is commonly used as it allows easier access to the distal lobe. Pig-tail catheter (5 or 6Fr.) is introduced inside the LAA via the access sheath to escort the access sheath into LAA ostium. Cine angiogram

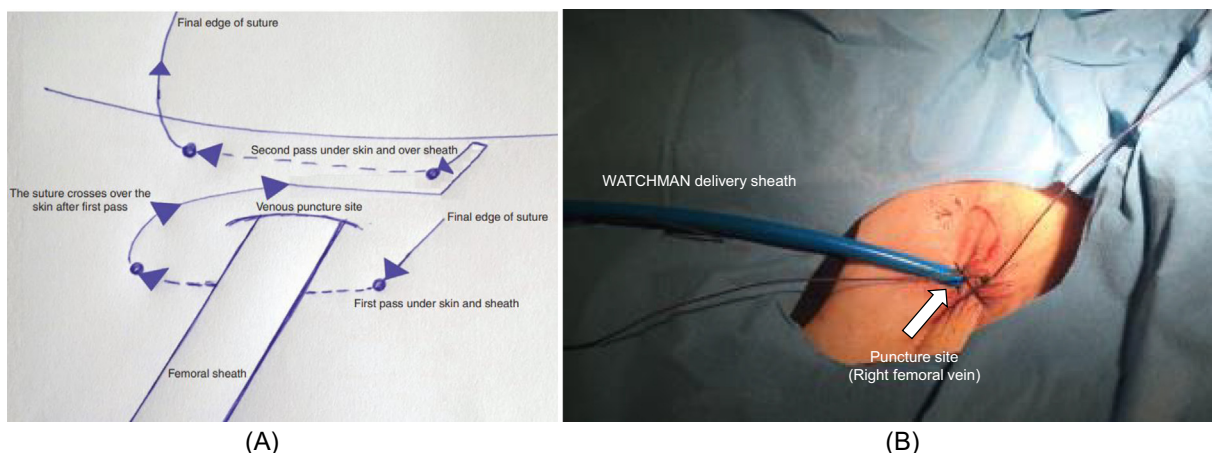


Fig. 5. “Figure of eight” technique. (A) How to suture “figure of eight”. (B) Picture of the right inguinal puncture site.

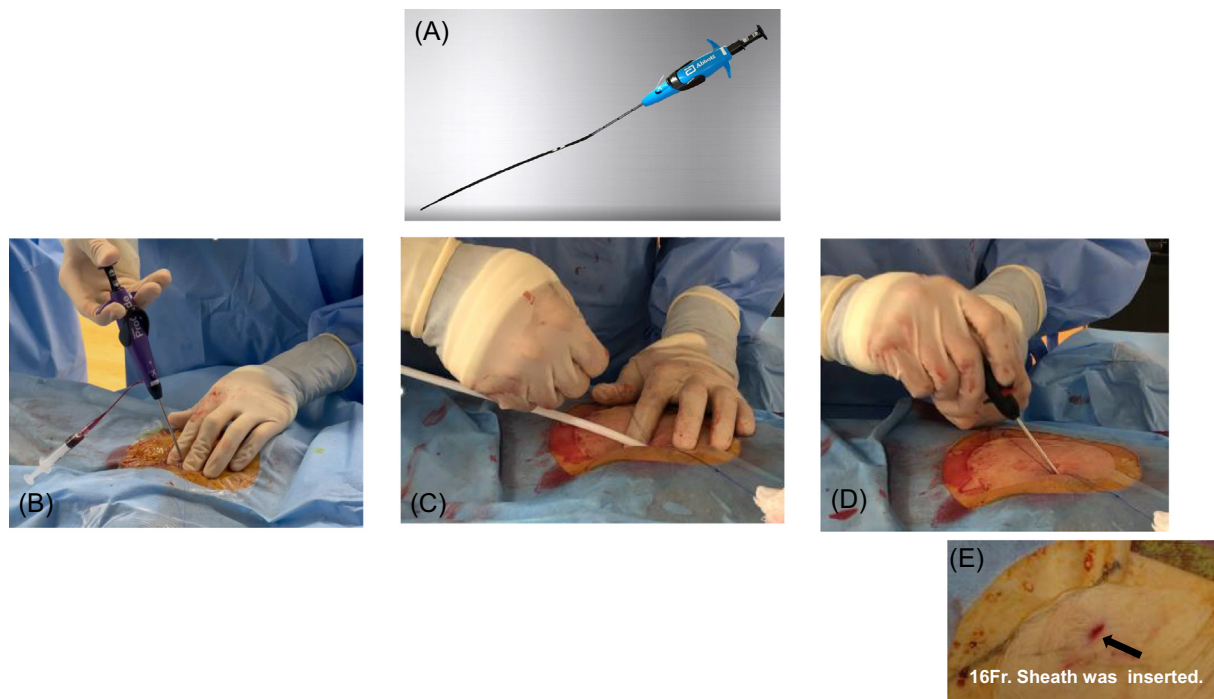


Fig. 6. Pre-close technique (16-Fr. sheath insertion). (A) Perclose ProStyle (suture-mediated closure system, Abbott Vascular, Temecula, CA, USA). (B) Pre-close before 16-Fr. sheath insertion. (C) After procedure, 16-Fr. sheath was removed. (D) Push the knot and trimming. (E) After hemostasis. Black arrow is 16Fr. sheath insertion point.

is performed at the right anterior oblique and caudal projection to conform LAA morphology by injection of contrast into LAA. The size of WATCHMAN FLX is determined by the maximum LAA ostium diameter under the condition of the LA pressure over 10 mm Hg.

The set of WATCHMAN FLX and device delivery catheter is inserted into the access sheath. After locking the set and sheath, the WATCHMAN FLX is ready to be deployed. WATCHMAN FLX is advanced into the tip of access sheath, and FLX deployment is accomplished by slowly advancing the delivery catheter and/or unsheathing the access sheath under fluoroscopy and 2-dimensional trans-esophageal echocardiography at 45 and 135°. WATCHMAN FLX can be partially or fully recaptured and redeployed if the position of device is not acceptable.

It is necessary to check the four points “PASS criteria” before completely releasing the device. P: optimal position, A: checking anchor to the LAA by tug test, S: sealing by FLX, and S: enough sizing from 10 % to 30 % of compression rate. The FLX is released completely by rotating the delivery catheter knob after confirming the PASS criteria.

Anti-thrombotic therapy after WATCHMAN implantation

A multitude of anti-thrombotic strategies after WATCHMAN implantation have been proposed, including warfarin, DOAC, aspirin monotherapy, DAPT, and no-therapy.

The duration and type of anti-thrombotic therapy after LAAC is initially based on the original clinical trials. From the PROTECT-AF and PREVAIL study, anti-coagulation with warfarin was used for 45 days after device implantation, followed by DAPT with aspirin and clopidogrel for 6 months, and then by aspirin alone for a lifetime [22,23].

The safety and efficacy of DAPT after LAAC was investigated for NVAf patients with contraindication for warfarin in the ASAP study [35]. DAPT with aspirin and clopidogrel was tested for 6 months after device implantation in 150 patients. The annual incidence of ischemic stroke was 1.7 %, suggesting the effectiveness of DAPT for contraindication of OAC. In the EVOLUTION trial [36], 27 % of patients were treated with warfarin/DOAC at discharge, 60 % with DAPT, 7 % received single antiplatelet therapy, and 8 % without any anti-thrombotic drugs. There

was no association between ischemic stroke or device-related thrombosis and the assigned types of medical therapy.

Søndergaard and colleagues compared anti-platelet therapy with anti-coagulation therapy using PROTECT-AF, PREVAIL, CAP, ASAP, and EVOLUTION trials [37]. There were no significant differences in the incidence of major bleeding or thromboembolism between anti-coagulation therapy and anti-platelet therapy (between 91 % DAPT and 9 % single anti-platelet therapy). However, device-related thrombosis was more common in the anti-platelet therapy group even though thromboembolic events were not reported.

In the PINNACLE FLX trial, the efficacy and safety of the WATCHMAN FLX device were investigated [30]. In this trial, DOACs were used as anti-coagulation after the procedure in all participants. The anti-coagulation regimen consisted of using DOAC and aspirin for 45 days, followed by DAPT until 6 months, and aspirin alone for a lifetime after that. Only aspirin therapy was achieved in 85.9 % of the patients at 12 months.

Recently, after LAAC using WATCHMAN, half-dose DOAC therapy significantly reduced the risk of thromboembolic and major bleeding events compared with standard anti-platelet therapy in patients with high risk of embolism and bleeding [38].

The methodology of anti-thrombotic therapy after WATCHMAN implantation is controversial. There are various types of anti-thrombotic therapy after device implantation. From previous clinical studies, the optimal regimen following device implantation can be variable and should be tailored to each patient. Of note, the optimal duration of using anti-thrombotic therapy is another continuing area of research.

Complications

Complications during LAAC are related to A) device access from femoral vein to fossa ovalis, B) device implantation, and C) device-related thrombosis. For device access complications, the use of body surface echo-guided puncture and hemostatic devices can reduce puncture site complications.

Pericardial effusion

Pericardial effusion is serious if hemodynamic compromise was present. Serious pericardial effusion was noted in 5.2 % in PROTECT-AF [22]. Several notable maneuvers during the procedure have been hypothesized to cause effusions. These are trans-septal puncture, manipulation of guidewire, delivery sheath in the LAA, and device deployment process.

Recently, learning curve of the operators, review of procedure, and especially, next-generation device on the market, the rate of pericardial effusions have dramatically decreased. In the PINNACLE-FLX trial [29], the incidence of serious pericardial effusion was only 1.0 %. The next-generation WATCHMAN FLX with closed-tip design (Fig. 3) may offer a great safety during the implantation process.

Device embolization

This complication is rare but can be serious. The incidence of device embolization was 0.6 % in the PROTECT-AF trial [22]. If the device migrates into the aorta or left atrium, it can be removed by percutaneous technique, although not easily, but if it migrates into the left ventricle, most of them require surgical removal. WATCHMAN FLX has dual raw anchors which are designed to reduce the risk of embolization with optimal engagement in the LAA (Fig. 3). The PINNACLE-FLX trial using WATCHMAN FLX observed no device embolization [29].

Device-related thrombosis

One of the biggest adverse events associated with the LAAC is the thrombus formation on the device, which is easily detected by trans-esophageal echocardiography and computed tomography (Fig. 7). The incidence of device-related thrombosis was 2–5 % and found frequently from day 45 to 12 months after device implantation [39].

Several patient-related and procedure-related factors are reported to be related with device-related thrombosis. The risk factors include hyper-coagulability, pericardial effusion, renal insufficiency, deep device implantation, non-paroxysmal AF, low left ventricular ejection fraction, history of thromboembolism, and large size device [40,41]. An annual stroke rate in patients with device-related thrombosis is only 0.3 % per 100 patient-years [42]. However, compared to the patients without device-related thrombosis, their risk of stroke is four times higher.

Treatment consists of anti-coagulation with heparin or OAC [43]. Thrombus disappeared in 95 % of cases using heparin or OAC with a varied treatment duration (median 45 days) [44]. Further studies are underway to investigate optimal anti-thrombotic regimen post-device

implantation. Randomized clinical trials would help in establishing the best regimen in the future, but for now, based on the use of current data, tailor-made therapy considering each individual's bleeding and embolism risk are essential for each patient.

Future directions

According to an operator's experience and an improvement of procedural technique and device, complications in LAAC are dramatically decreasing. The PINNACLE FLX study showed only 0.5 % of the primary safety endpoint (death, ischemic stroke, systemic embolism, procedure-related events requiring surgery/intervention) and 98.8 % of implant success using the WATCHMAN FLX [37].

Based on such excellent data, the CHAMPION-AF trial has been initiated in many countries around the world, including Japan. This trial investigates the safety and efficacy of LAAC with WATCHMAN FLX compared with long-term administration of DOAC in patients with NVAF, irrespective of the tolerability to long-term anticoagulant therapy. The results of this study may lead to a paradigm shift in cardiogenic stroke prevention in NVAF patients.

Various forms of LAAC devices have been innovated in Europe and the USA. One of them, AMPLATZER AMULET was approved in the USA in 2021. This is a self-expanding device made of a braided nitinol mesh with two polyester patches sewn onto a distal lobe and a proximal disc connected by a short waist (Fig. 2C). Unlike WATCHMAN of “plug” design, AMULET has a distal “anchor” body and proximal “sealing” disc design. Because this “anchor and seal” approach can anchor the proximal part of LAA and seal the LAA ostium with the disc, AMULET may be superior to WATCHMAN for more complex LAA structures like multiple lobes and shallow lobes [45]. This device is expected to be introduced in Japan in the near future.

LAAC procedures are performed under general anesthesia with trans-esophageal echocardiography guidance. Trans-esophageal echocardiography allows multi-planar and 3-dimensional imaging, which can be essential for the assessment of the highly variable LAA. In the instruction for use of WATCHMAN, LAAC requires trans-esophageal echocardiography imaging at four different views to ensure acceptable position, anchor, seal, and compression (P.A.S.S. criteria) in Japan.

On the other hand, many patients who are referred to LAAC are elderly and have a high burden of comorbidities and high prevalence of frailty, avoidance of general anesthesia would plausibly yield several benefits. Additionally, trans-esophageal echocardiography is not without risk, especially in elderly and frail populations. A minimally invasive method to perform LAAC without general anesthesia and trans-esophageal echocardiography, intra-cardiac echocardiography-guided LAAC has been recently proposed [46]. However, at present, intra-

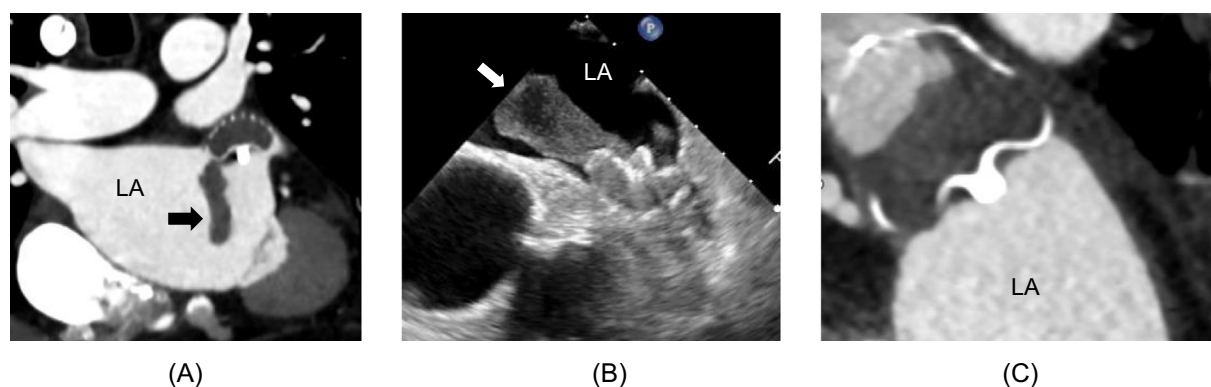


Fig. 7. Device-related thrombosis (DRT). (A) DRT detected by contrast-enhanced computed tomography. (B) DRT detected by trans-esophageal echocardiography. (C) DRT disappeared after treatment by heparin. LA, left atrium; arrow, thrombus.

cardiac echocardiography-guided LAAC has problems with the learning curve, intra-cardiac echocardiography technology such as the inability to create multi-planar and 3-dimensional images, and the lack of imaging standards at LAAC.

Conclusions

LAAC for patients with NVAF may be an alternative therapy to avoid cardiac embolism for high bleeding risk patients with contraindications to long-term OAC therapy. With the advent of WATCHMAN FLX, improved treatment techniques, and the operator's learning curve, LAAC has become safer. With the innovation of further devices, the introduction of minimally invasive procedures such as trans-esophageal echocardiography guided-LAAC, and the establishment of optimal anti-coagulant therapy to prevent thromboembolic events, LAAC may become the first choice for cardiogenic embolism prophylaxis in all patients with NVAF.

Declaration of competing interest

None.

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None.

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